ROLL NO: | | | | | | | | | | |

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INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

Diploma (CSE)

Semester: VI

SUBJECT CODE: DCS602

SOFTWARE ENGINEERING

Time: 03:00 Hrs. **Maximum Marks: 60**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. (12 Marks)

Q. 1. Answer the following questions: (1×6 = 6 Marks)

(i) Define Software Engineering.

(ii) List the different phases of the Software Development Life Cycle (SDLC).

(iii) State the purpose of requirement analysis in software development.

(iv) Name any two software development models.

(v) Mention two characteristics of good software.

(vi) Draw a simple diagram showing the Waterfall Model.

Q. 2. Answer the following questions: (1×6 = 6 Marks)

(i) Paraphrase the meaning of "software process model" in your own words.

(ii) Clarify the difference between verification and validation in software engineering.

(iii) Translate the concept of Agile methodology into a simple daily life example.

(iv) Represent the V-Model of software development with a basic diagram.

(v) Clarify the role of a Software Requirements Specification (SRS) document.

(vi) Paraphrase the advantages of using prototyping in software development.

SECTION – B

NOTE:Attempt all questions. (6×2×2 = 24 Marks)

Q.3. (i)(a) Draw a Data Flow Diagram (DFD) for an Online Banking System showing at least two levels of decomposition.

OR

(b) Solve a problem to estimate the cost of a software project using the Basic COCOMO model, given required parameters.

(ii)(a) Modify the Waterfall Model to accommodate iterative development and explain how it changes the software process.

OR

(b) Relate the phases of the SDLC to a real-world example such as building an e-commerce website.

Q.4. (i) (a) Differentiate between Black Box Testing and White Box Testing with examples.

OR

(b) Outline the key activities involved in Software Configuration Management (SCM).

(ii)(a) Select an appropriate software development model (Waterfall, Spiral, Agile) for developing a mobile application and justify your selection.

OR

(b) Integrate risk analysis into the project planning process and explain its importance with an example.

SECTION – C

NOTE: Attempt all questions. (12×2 = 24 Marks)

Q. 5. (a) Judge the effectiveness of Agile methodology compared to the Waterfall model for a project with rapidly changing requirements. Highlight the strengths and weaknesses of each model.

OR

(b) Evaluate the significance of Software Quality Assurance (SQA) activities in ensuring the reliability and success of software products. Provide examples to support your answer.

Q. 6. (a) Plan and design a software requirement specification (SRS) document for an Online Food Delivery System. Include the scope, functional requirements, non-functional requirements, and use-case diagrams.

OR

(b) Generate a detailed design for a Hospital Management System. Classify the system modules (like patient management, appointment scheduling, billing, etc.) and create a high-level architecture diagram explaining their interaction.